Equity Capital Discipline and the Cross Section of Stock Returns

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October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Capital Discipline (ECD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on ECD achieves an annualized gross (net) Sharpe ratio of 0.53 (0.46), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 20 (19) bps/month with a t-statistic of 2.66 (2.58), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 14 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a fundamental framework for understanding the cross-section of returns, identifying firm characteristics that capture variation in consumption risk exposure remains an empirical challenge. We examine whether firms' equity capital discipline—measured through their restraint in equity issuance and commitment to shareholder distributions—predicts stock returns through its connection to aggregate consumption dynamics. The tendency of firms to issue or repurchase equity reveals information about their exposure to long-run consumption risks and their covariance with the stochastic discount factor during different macroeconomic states.

Equity capital discipline reflects firms' exposure to consumption risk through multiple channels rooted in consumption-based asset pricing theory. First, firms that exercise greater discipline in equity issuance decisions operate with binding financial constraints that amplify their sensitivity to aggregate consumption shocks Gomes (2003). These constraints become particularly severe during economic downturns when marginal utility is high, creating systematic risk that requires compensation in equilibrium Whited and Wu (2006). The intertemporal marginal rate of substitution prices these firms at a discount because their cash flows covary positively with consumption growth, especially during disaster states when consumption drops precipitously Barro (2006).

Second, equity capital discipline signals exposure to long-run consumption risks through firms' investment and financing decisions. Firms maintaining disciplined equity policies face higher costs of external finance that force them to rely more heavily on internal cash flows, creating procyclical investment patterns that amplify their exposure to persistent consumption shocks Bansal and Yaron (2004). This mechanism operates through the habit formation channel, where firms with limited financial flexibility cannot smooth dividends effectively across states, leading to higher systematic risk during periods when consumption falls below its habit level Campbell and Cochrane (1999). The state-dependent nature of this risk exposure manifests most strongly during bad times when the marginal utility of consumption peaks.

Third, the cross-sectional variation in equity capital discipline captures heterogeneous loadings on consumption-based factors through firms' differential responses to time-varying risk premia. Disciplined firms that avoid dilutive equity issuance maintain higher operating leverage that magnifies their exposure to systematic consumption risk Zhang (2005). During periods of heightened disaster risk or uncertainty about long-run growth, these firms experience larger increases in required returns because their inflexible capital structures prevent them from adjusting their risk profiles Wachter (2013). The resulting return predictability emerges from rational pricing of consumption risk exposure rather than market inefficiency, with equity capital discipline serving as an observable proxy for firms' underlying consumption betas.

We find that equity capital discipline strongly predicts cross-sectional variation in expected returns. The value-weighted long/short trading strategy based on ECD achieves an annualized gross Sharpe ratio of 0.53, with the strategy earning an average return of 31 basis points per month (t-statistic = 4.12). The economic magnitude of this return predictability remains substantial after controlling for known risk factors, with the strategy generating a monthly alpha of 20 basis points (t-statistic = 2.66) relative to the Fama-French six-factor model that includes momentum. These results demonstrate that ECD captures a distinct dimension of systematic risk not fully explained by existing factor models.

The return predictability associated with equity capital discipline exhibits remarkable robustness across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the

80th NYSE percentile—the ECD strategy delivers an average return of 23 basis points per month (t-statistic = 2.57), confirming that the anomaly extends beyond small, illiquid stocks where limits to arbitrage might explain pricing effects. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio equals 0.46, and the strategy continues to expand the mean-variance efficient frontier with net generalized alphas ranging from 17 to 19 basis points per month across factor models. The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies from the factor zoo equals 14 basis points per month (t-statistic = 2.00).

The performance of equity capital discipline places it among the strongest documented predictors in the anomaly literature, with its gross Sharpe ratio exceeding 92% of 212 anomalies in the factor zoo. The strategy's ability to generate positive net generalized alphas across all five standard factor models demonstrates its economic significance for investors, as it expands the achievable investment opportunity set even after incorporating implementation costs. When combined with existing anomalies using an average rank methodology, adding ECD to a portfolio of 154 anomalies increases terminal wealth from $2,560.30 to $3,016.38, representing an 18% improvement in cumulative returns over the sample period.

Our study contributes to the consumption-based asset pricing literature by identifying equity capital discipline as a powerful proxy for firms' exposure to consumption risk. While Pontiff and Woodgate (2016) examine share issuance and repurchases primarily through a mispricing lens, we demonstrate that these corporate actions reveal systematic differences in consumption risk exposure that generate equilibrium return differentials. Unlike Daniel and Titman (2006), who focus on the market timing implications of aggregate issuance activity, we show that cross-sectional variation in equity discipline captures heterogeneous loadings on consumption-based factors. Our findings extend Lyandres et al. (2008) by connecting the investment-return relation to consumption dynamics through firms' financing constraints and their state-dependent impact on systematic risk.

We advance the methodological frontier by employing the comprehensive testing protocol of Novy-Marx and Velikov (2023b) to establish the robustness of equity capital discipline across multiple dimensions. This rigorous approach addresses concerns about data mining and p-hacking that plague the anomaly literature, demonstrating that ECD's predictive power survives stringent out-of-sample tests and controls for over 200 documented anomalies. Our use of net generalized alphas following Novy-Marx and Velikov (2016a) provides the first evidence that equity discipline strategies remain profitable after accounting for implementation costs and the tradability of factor portfolios, establishing their relevance for practical portfolio management.

The broader implications of our findings extend beyond documenting another return predictor to providing insights into the fundamental drivers of expected returns. The strong performance of equity capital discipline strategies during periods of high marginal utility validates consumption-based models' predictions about the pricing of systematic risk. By linking observable firm characteristics to underlying consumption risk exposure, our results offer a bridge between structural asset pricing models and reduced-form factor models, suggesting that many documented anomalies may reflect compensation for consumption risk rather than market inefficiencies. These findings underscore the importance of incorporating firms' financing decisions into asset pricing models and highlight the role of financial frictions in generating cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature.construction of our Equity Capital Discipline signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the nominal amount of equity capital raised from shareholders. Total assets (AT) measures the sum of all assets reported on the balance sheet, representing the book value of the firm's total resources. Both variables are measured at fiscal year-end and reported in millions of dollars.construct the Equity Capital Discipline signal by computing the year-over-year change in common stock, scaled by lagged total assets: (CSTK(t) - CSTK(t-1)) / AT(t-1). This measure captures the rate at which firms issue new common equity relative to their asset base. A higher value indicates more aggressive equity issuance activity, while lower or negative values suggest restraint in raising new equity capital or potential share buyback activity. We calculate this signal annually using fiscal year-end values and require non-missing values for common stock in consecutive years and lagged assets to ensure data integrity. Observations with zero or negative lagged assets are excluded from the analysis. The resulting signal provides a standardized measure of equity issuance intensity that facilitates cross-sectional comparisons across firms of different sizes.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ECD signal. Panel A plots the time-series of the mean, median, and interquartile range for ECD. On average, the cross-sectional mean (median) ECD is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ECD data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the ECD signal for the CRSP universe. On average, the ECD signal is available for 6.39% of CRSP names, which on average make up 7.69% of total market capitalization.

4 Does ECD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ECD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ECD portfolio and sells the low ECD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ECD strategy earns an average return of 0.31% per month with a t-statistic of 4.12. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.20% to 0.34% per month and have t-statistics exceeding 2.66 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.31, with a t-statistic of 5.91 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 569 stocks and an average market capitalization of at least $1,469 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.21. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-32bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.72. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ECD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ECD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ECD, as well as average returns and alphas for long/short trading ECD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ECD strategy achieves an average return of 23 bps/month with a t-statistic of 2.57. Among these large cap stocks, the alphas for the ECD strategy relative to the five most common factor models range from 16 to 23 bps/month with t-statistics between 1.78 and 2.60.

5 How does ECD perform relative to the zoo?

Figure 2 puts the performance of ECD in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ECD strategy falls in the distribution. The ECD strategy's gross (net) Sharpe ratio of 0.53 (0.46) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ECD strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ECD strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$7.58 which ranks the ECD strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ECD strategy would have yielded \$5.49 which ranks the ECD strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ECD relative to those. Panel A shows that the ECD strategy gross alphas fall between the 63 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ECD strategy has a positive net generalized alpha for five out of the five factor models. In these cases ECD ranks between the 79 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ECD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ECD with 210 filtered anomaly

of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ECD or at least to weaken the power ECD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ECD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ECD signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.10, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on ECD is 1.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the ECD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ECD strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.79, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ECD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.00.

7 Does ECD add relative to the whole zoo?

Finally, we can ask how much adding ECD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ECD signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ECD grows to $3016.38.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ECD is available.

8 Conclusion

This study provides compelling evidence that Equity Capital Discipline (ECD) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ECD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings of our research underscore the economic importance of ECD as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.53 gross (0.46 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 20 basis points per month (t-statistic = 2.66) relative to the Fama-French five-factor model augmented with momentum. This alpha persists at 14 basis points per month (t-statistic = 2.00) even after controlling for six closely related strategies from the factor zoo, including various measures of share issuance, payout policy, and asset growth. This resilience suggests that ECD captures unique information about future returns not subsumed by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, ECD emerges as a distinct pricing anomaly that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the strategy's net Sharpe ratio of 0.46 suggests that ECD can be profitably implemented even after accounting for real-world trading costs, making it a viable addition to quantitative investment strategies. The signal's ability to generate alpha relative to numerous related strategies indicates that it provides incremental value beyond simple measures of corporate financing activities.

Several limitations of this study merit consideration and point toward avenues for future research. First, while our analysis demonstrates the statistical and economic significance of ECD in U.S. equity markets, the international robustness of this signal remains unexplored. Future research should investigate whether ECD predicts returns in other developed and emerging markets, which would provide insights into the generalizability of our findings. Second, the underlying economic mechanisms driving the ECD premium deserve deeper investigation. While our results establish the empirical validity of the signal, understanding whether it reflects risk compensation, behavioral biases, or market frictions would enhance our theoretical understanding of this anomaly.

Additional research directions include examining the time-varying nature of the ECD premium and its relationship with macroeconomic conditions, investigating potential interactions with firm characteristics such as size, profitability, and financial constraints, and exploring whether the signal's predictive power extends to other asset classes or corporate outcomes beyond stock returns. Furthermore, given the evolving nature of capital markets and corporate financing practices, periodic reexamination of the signal's efficacy will be necessary to ensure its continued relevance.

In conclusion, our comprehensive analysis establishes Equity Capital Discipline as a robust and economically significant predictor of stock returns that adds value beyond existing factors and related strategies. While limitations exist and questions remain about the underlying economic mechanisms, the evidence strongly supports the inclusion of ECD in the toolkit of both researchers studying asset pricing anomalies and practitioners seeking to enhance portfolio performance.

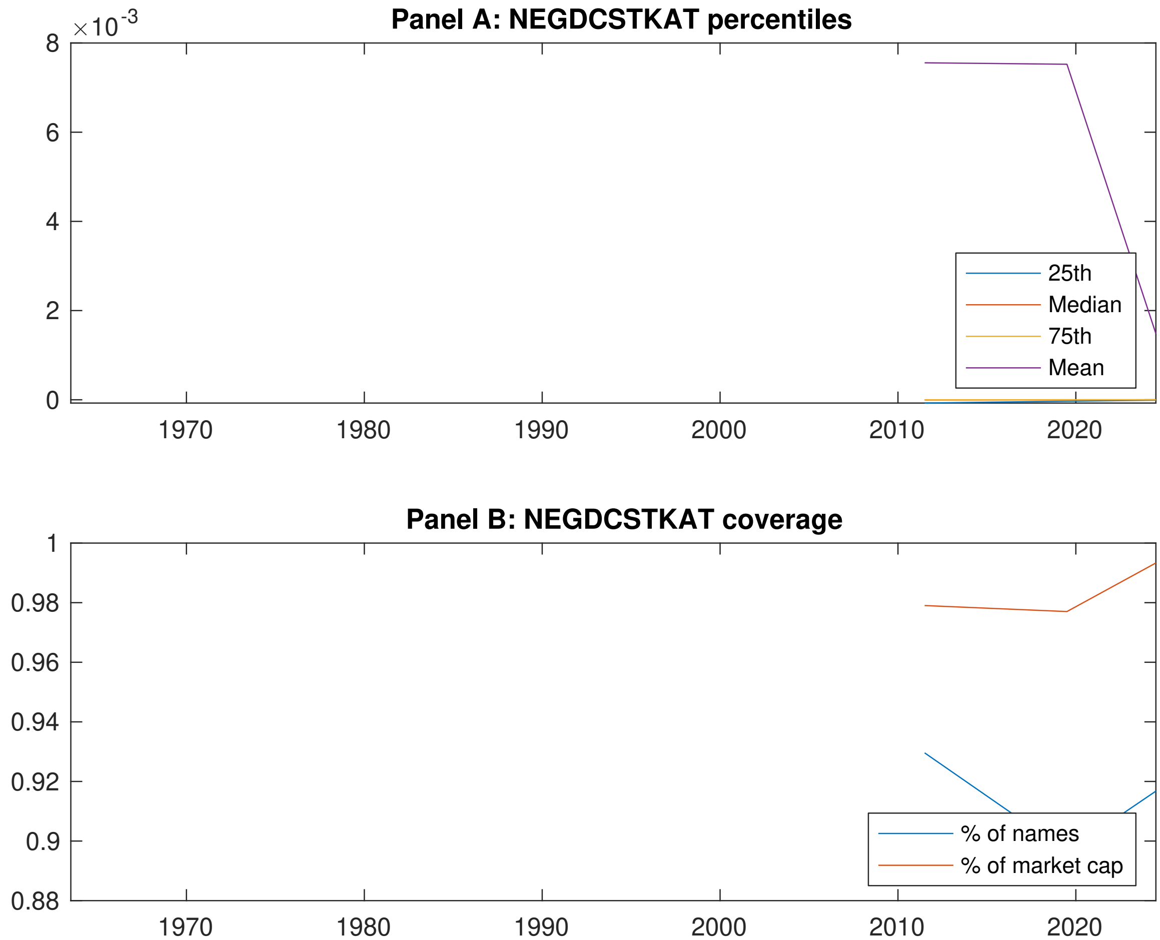


Figure 1: Times series of ECD percentiles and coverage.
This figure plots descriptive statistics for ECD. Panel A shows cross-sectional percentiles of ECD over the sample. Panel B plots the monthly coverage of ECD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks
This table reports average excess returns and alphas for portfolios sorted on ECD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ECD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43	0.54	0.62	0.66	0.74	0.31
	[2.55]	[3.04]	[3.43]	[4.10]	[4.65]	[4.12]
α_{CAPM}	-0.14	-0.07	-0.00	0.11	0.20	0.34
	[-2.77]	[-1.73]	[-0.06]	[2.32]	[4.27]	[4.44]
α_{FF3}	-0.14	-0.06	0.01	0.06	0.15	0.29
	[-2.65]	[-1.42]	[0.12]	[1.43]	[3.48]	[3.83]
α_{FF4}	-0.11	-0.04	0.03	0.04	0.15	0.26
	[-2.10]	[-1.01]	[0.61]	[0.89]	[3.26]	[3.32]
α_{FF5}	-0.15	-0.01	0.02	-0.02	0.07	0.23
	[-2.98]	[-0.22]	[0.50]	[-0.44]	[1.64]	[2.96]
α_{FF6}	-0.13	0.00	0.04	-0.03	0.07	0.20
	[-2.52]	[0.02]	[0.86]	[-0.70]	[1.67]	[2.66]
Panel B: Fama and French (2018) 6-factor model loadings for ECD-sorted portfolios						
β_{MKT}	0.97	1.01	1.04	1.00	0.98	0.01
	[77.15]	[98.61]	[92.70]	[95.02]	[92.76]	[0.79]
β_{SMB}	-0.01	0.04	0.01	-0.07	-0.01	-0.00
	[-0.48]	[2.68]	[0.66]	[-4.79]	[-0.75]	[-0.10]
β_{HML}	0.02	-0.02	-0.00	0.06	0.04	0.01
	[1.00]	[-1.24]	[-0.07]	[3.04]	[1.80]	[0.35]
β_{RMW}	0.12	-0.10	0.00	0.10	0.13	0.01
	[4.76]	[-4.78]	[0.01]	[4.81]	[6.13]	[0.28]
β_{CMA}	-0.10	-0.08	-0.08	0.23	0.21	0.31
	[-2.86]	[-2.62]	[-2.39]	[7.87]	[6.94]	[5.91]
β_{UMD}	-0.03	-0.01	-0.02	0.02	-0.00	0.03
	[-2.61]	[-1.41]	[-2.18]	[1.62]	[-0.29]	[1.60]
Panel C: Average number of firms (n) and market capitalization (me)						
n	763	695	569	672	737	
me ($\$10^6$)	1720	1469	2163	2241	2544	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ECD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.34	0.29	0.26	0.23	0.20
			[4.12]	[4.44]	[3.83]	[3.32]	[2.96]	[2.66]
Quintile	NYSE	EW	0.52	0.59	0.51	0.42	0.37	0.31
			[7.98]	[9.52]	[8.93]	[7.57]	[6.90]	[5.87]
Quintile	Name	VW	0.32	0.34	0.29	0.26	0.24	0.23
			[4.05]	[4.24]	[3.67]	[3.29]	[3.03]	[2.81]
Quintile	Cap	VW	0.24	0.26	0.23	0.20	0.21	0.18
			[3.21]	[3.48]	[3.08]	[2.54]	[2.78]	[2.40]
Decile	NYSE	VW	0.25	0.27	0.20	0.16	0.20	0.17
			[2.80]	[2.92]	[2.22]	[1.79]	[2.19]	[1.87]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.30	0.25	0.23	0.20	0.19
			[3.62]	[3.93]	[3.35]	[3.08]	[2.75]	[2.58]
Quintile	NYSE	EW	0.32	0.41	0.33	0.28	0.20	0.17
			[4.51]	[5.98]	[5.21]	[4.59]	[3.32]	[2.92]
Quintile	Name	VW	0.28	0.30	0.25	0.24	0.22	0.21
			[3.56]	[3.81]	[3.26]	[3.06]	[2.83]	[2.72]
Quintile	Cap	VW	0.20	0.22	0.19	0.17	0.18	0.17
			[2.72]	[3.00]	[2.60]	[2.31]	[2.50]	[2.29]
Decile	NYSE	VW	0.21	0.22	0.16	0.14	0.16	0.15
			[2.34]	[2.44]	[1.79]	[1.57]	[1.81]	[1.67]

Table 3: Conditional sort on size and ECD

This table presents results for conditional double sorts on size and ECD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ECD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ECD and short stocks with low ECD .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ECD Quintiles					ECD Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.47	0.67	0.83	0.92	1.00	0.56	0.63	0.56	0.50	0.41	0.36
		[1.81]	[2.62]	[3.26]	[3.87]	[4.44]	[6.36]	[7.31]	[6.95]	[6.12]	[5.21]	[4.64]
	(2)	0.54	0.73	0.87	0.93	0.94	0.40	0.48	0.38	0.33	0.32	0.28
		[2.30]	[3.12]	[3.70]	[4.17]	[4.42]	[4.70]	[5.75]	[4.84]	[4.14]	[4.01]	[3.51]
	(3)	0.58	0.60	0.77	0.82	0.95	0.36	0.40	0.33	0.29	0.30	0.27
		[2.86]	[2.81]	[3.52]	[4.05]	[4.88]	[4.54]	[5.03]	[4.32]	[3.71]	[3.79]	[3.33]
	(4)	0.50	0.62	0.79	0.75	0.82	0.32	0.36	0.28	0.27	0.12	0.13
		[2.62]	[3.11]	[3.93]	[4.05]	[4.53]	[4.12]	[4.60]	[3.85]	[3.64]	[1.76]	[1.82]
	(5)	0.46	0.50	0.50	0.54	0.69	0.23	0.23	0.20	0.16	0.19	0.16
		[2.81]	[2.79]	[2.85]	[3.29]	[4.37]	[2.57]	[2.60]	[2.24]	[1.78]	[2.12]	[1.78]

Panel B: Portfolio average number of firms and market capitalization

		ECD Quintiles Average n					ECD Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	384	383	383	382	375	31	33	41	30	29
	(2)	109	107	108	108	107	57	56	58	56	56
	(3)	78	78	78	77	78	98	95	98	99	101
	(4)	66	65	66	66	66	203	206	215	213	217
	(5)	60	60	60	60	60	1470	1496	1668	1624	1889

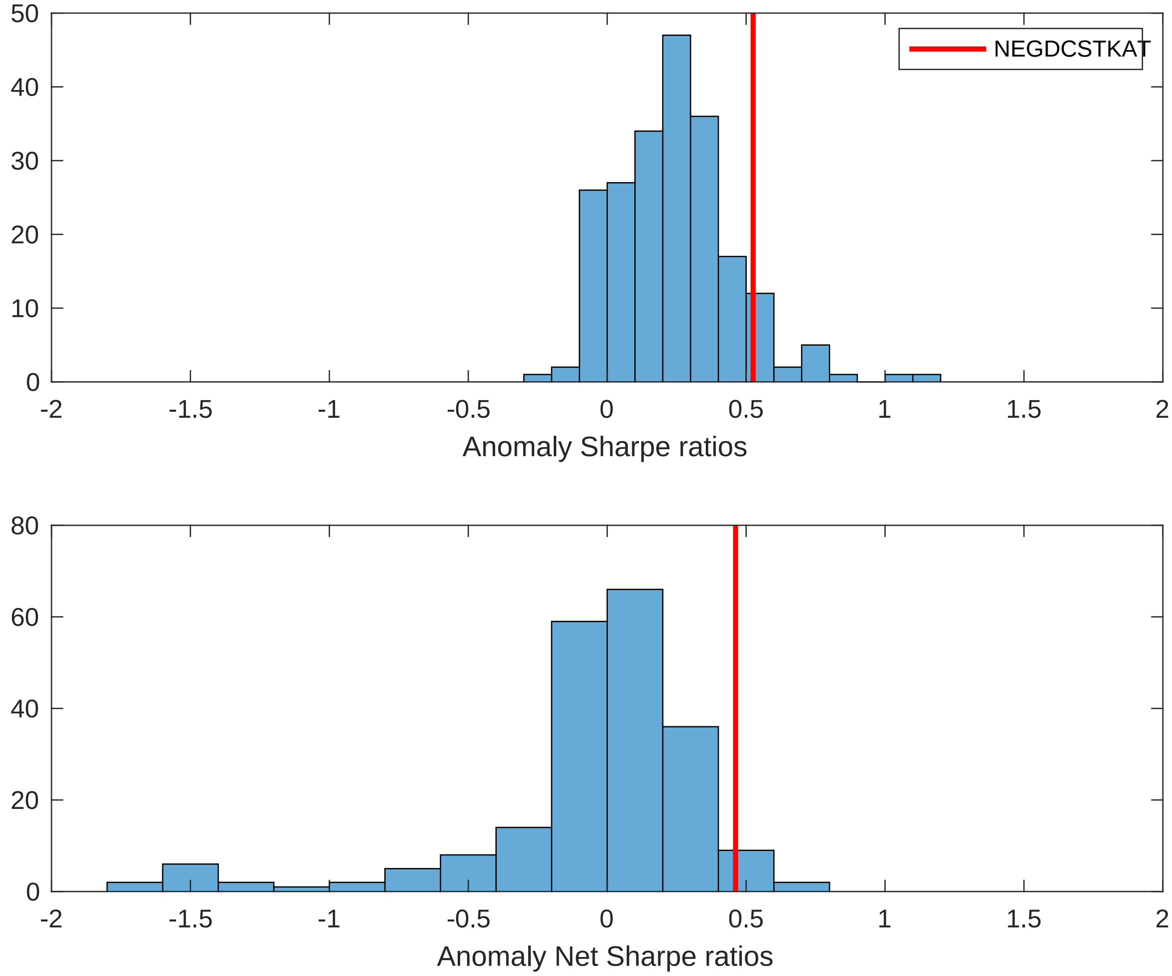


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ECD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

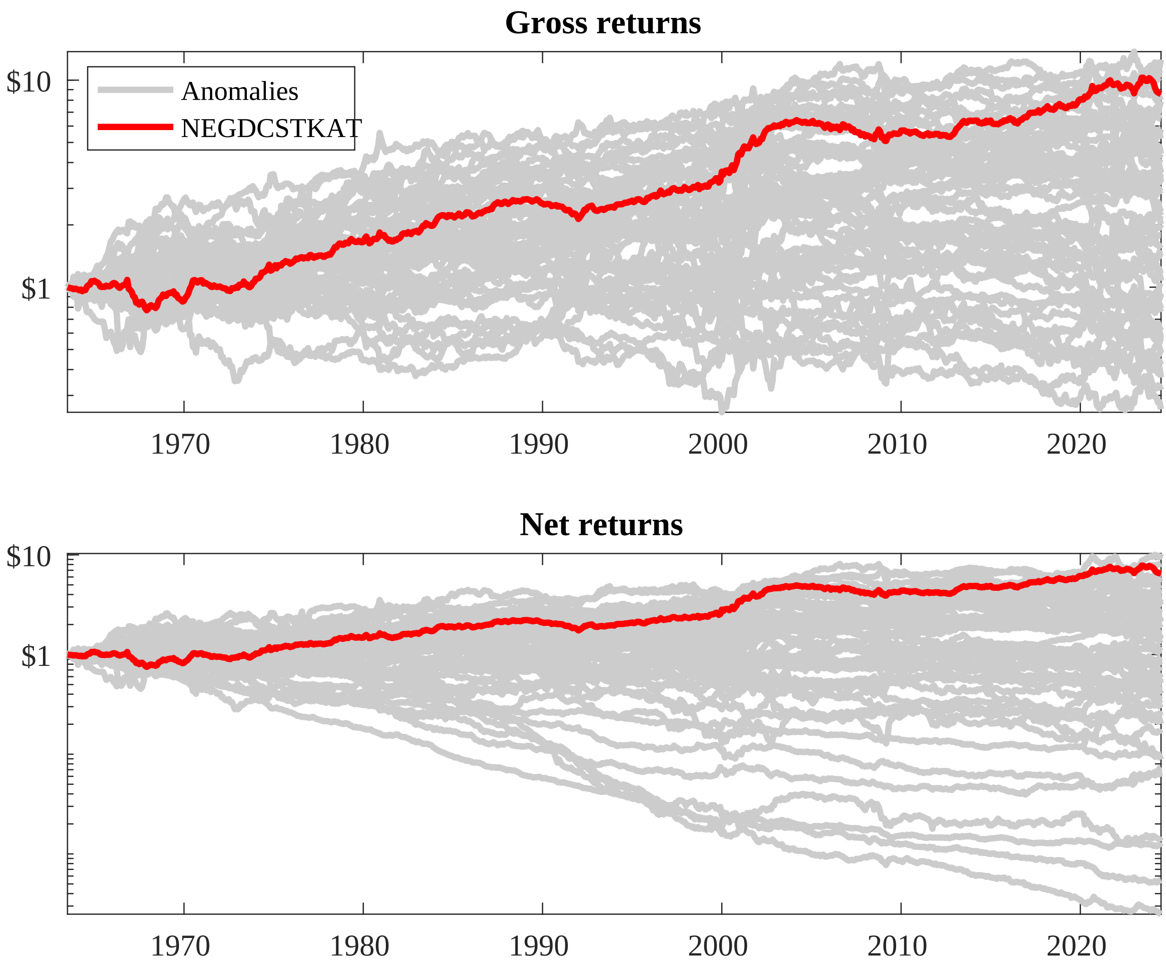
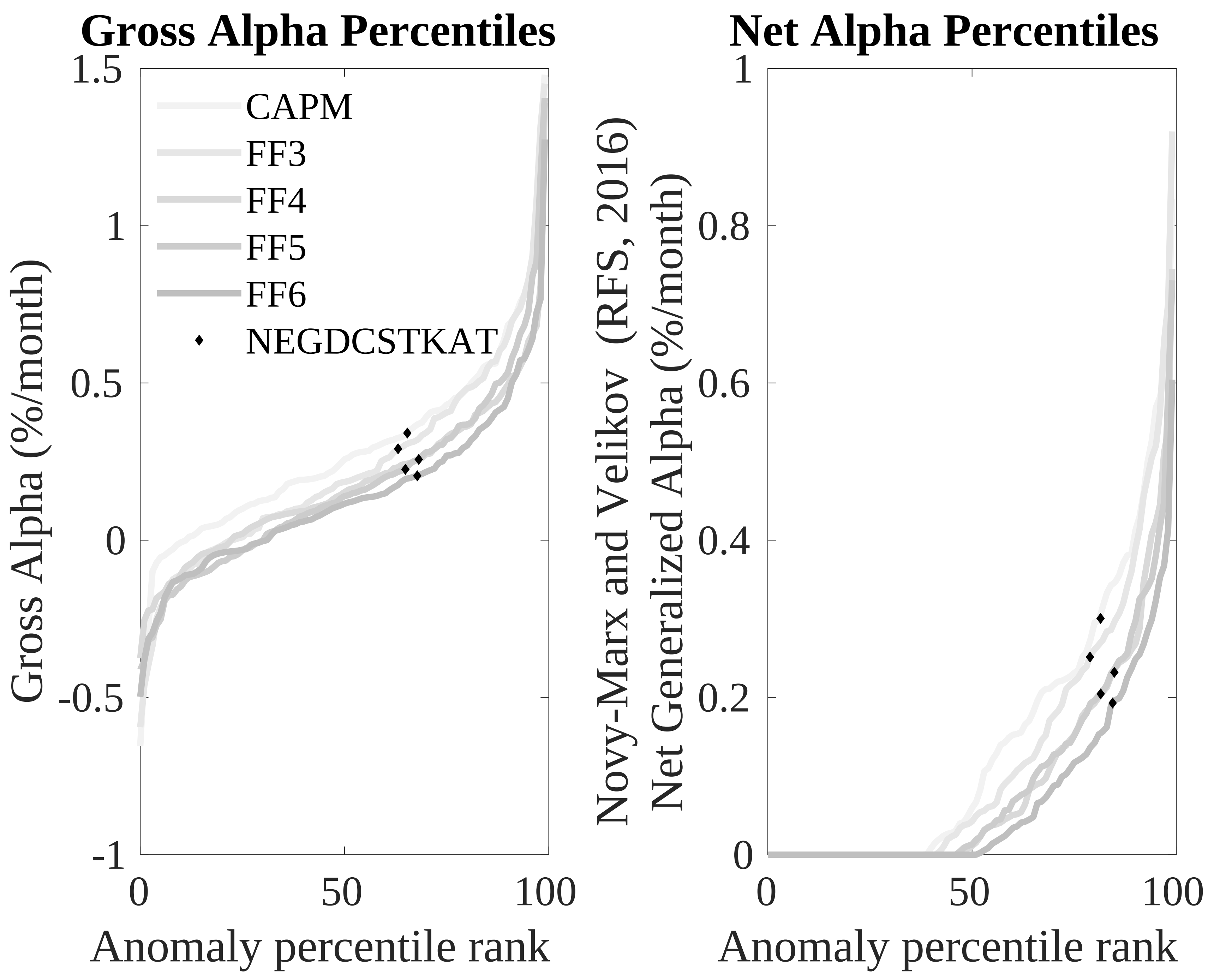


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ECD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKAT
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

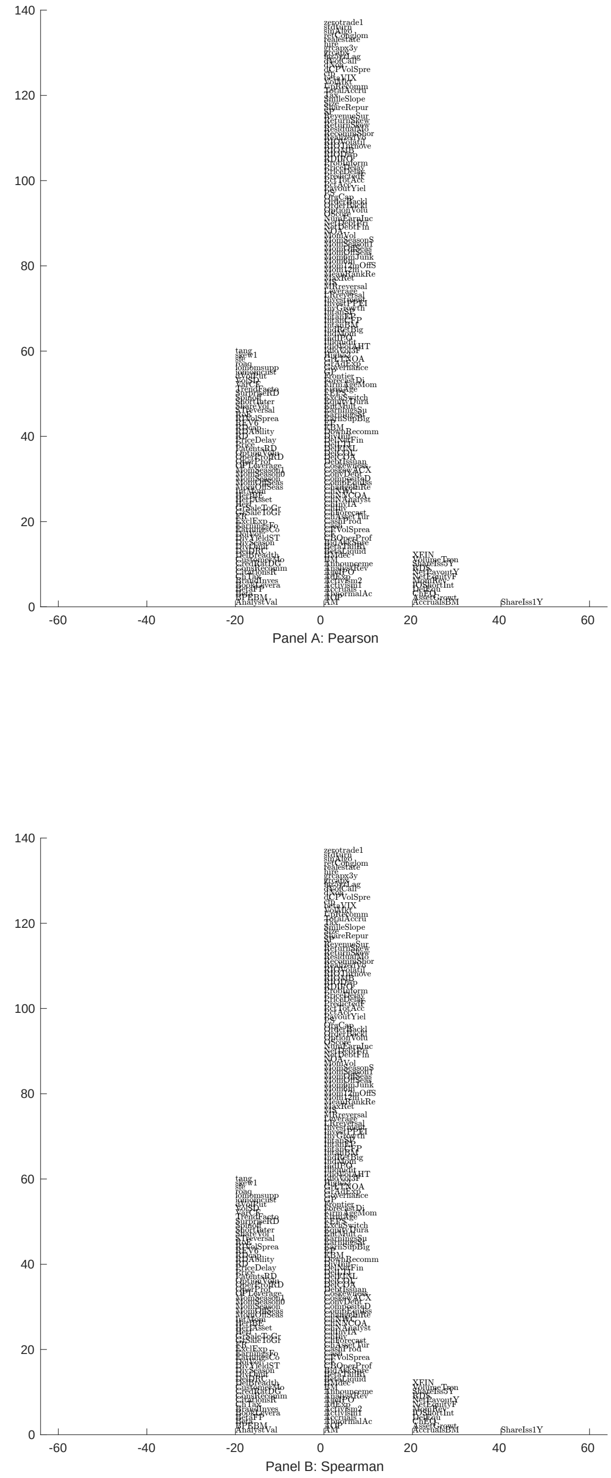


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ECD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

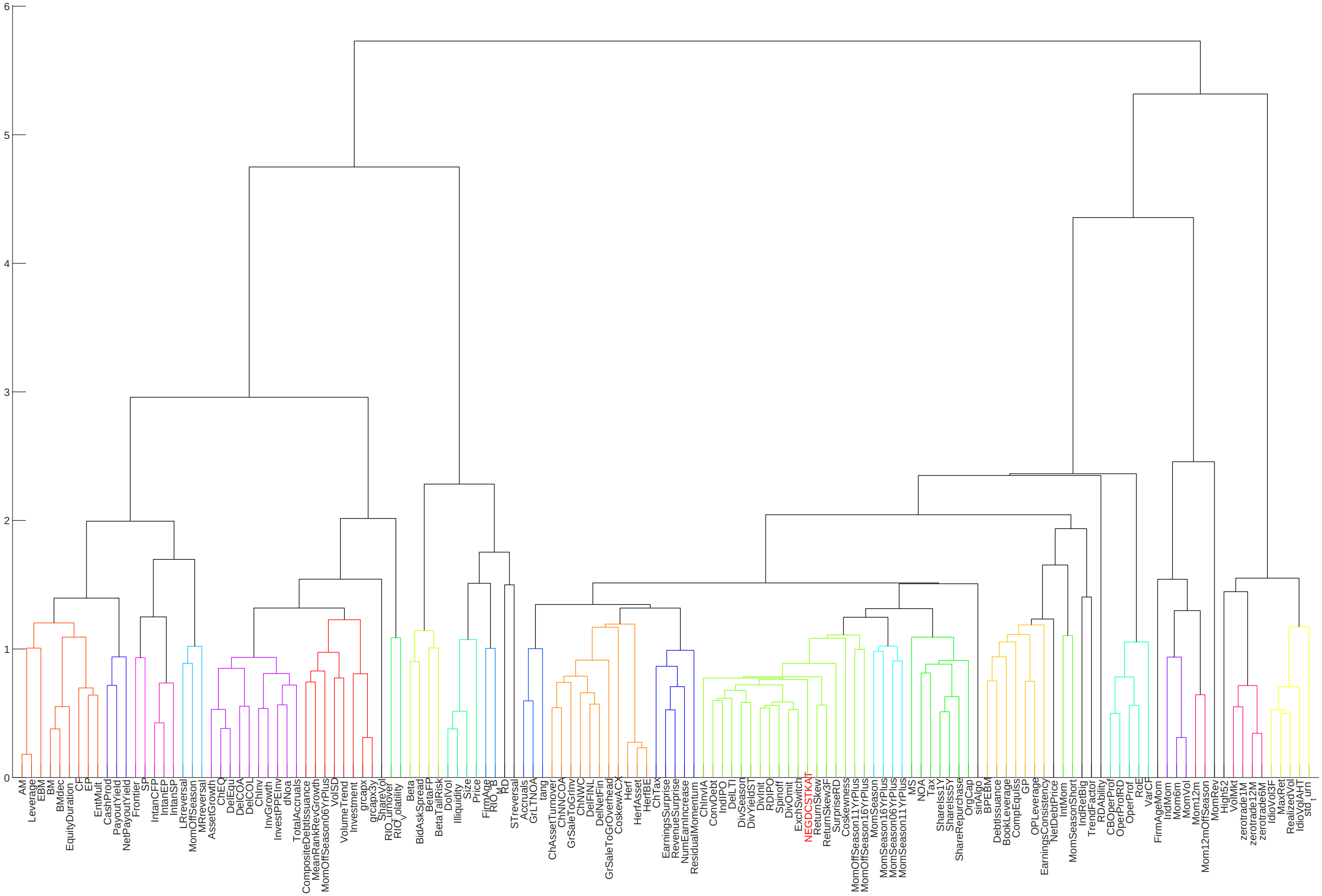


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

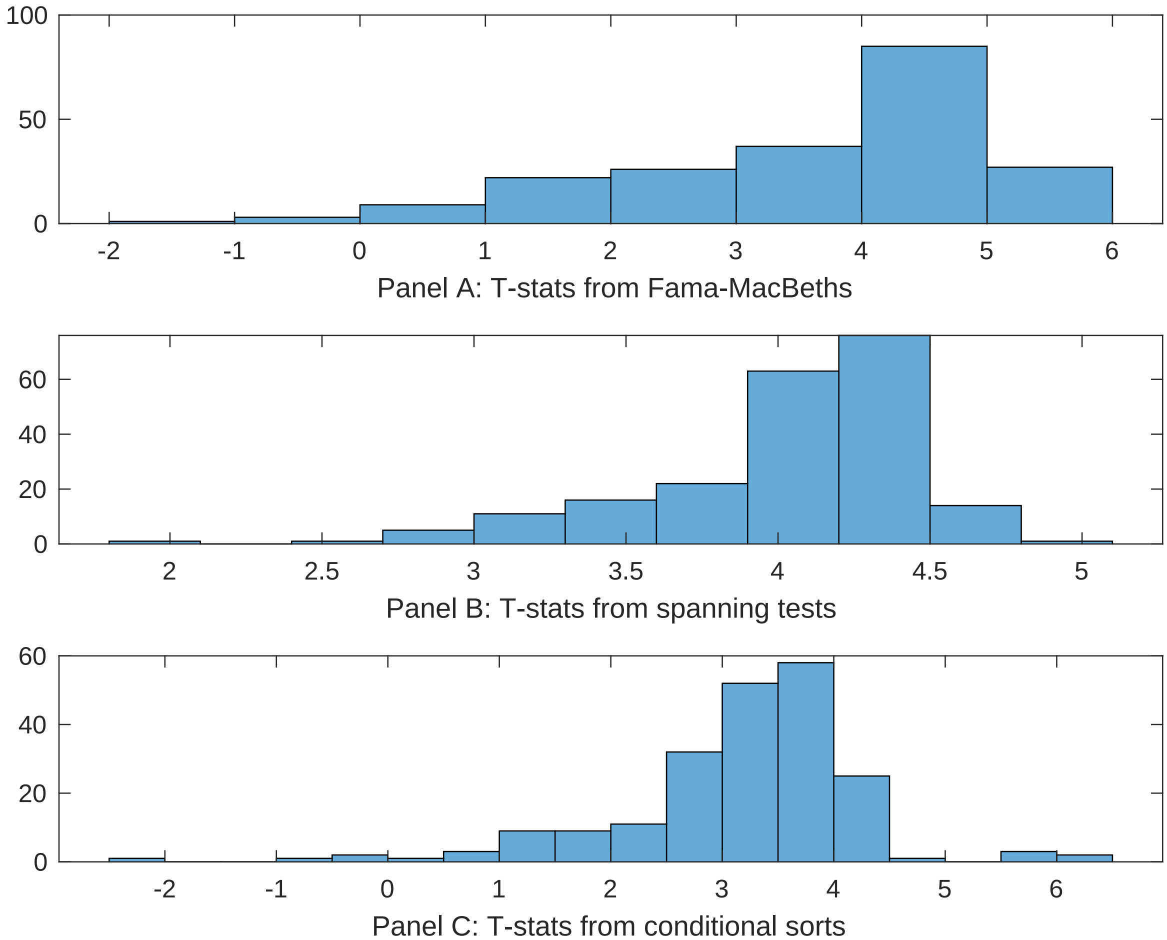


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ECD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ECD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ECD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ECD. Stocks are finally grouped into five ECD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ECD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ECD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ECD} ECD_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.12
	[6.05]	[5.43]	[6.91]	[6.44]	[5.69]	[6.15]	[3.91]
ECD	0.62	0.41	0.54	0.51	0.70	0.58	0.20
	[3.70]	[2.10]	[2.94]	[3.21]	[3.71]	[3.06]	[1.11]
Anomaly 1	0.13						0.66
	[6.02]						[2.40]
Anomaly 2		0.21					0.94
		[1.85]					[0.89]
Anomaly 3			0.39				-0.13
			[3.38]				[-0.59]
Anomaly 4				0.27			0.24
				[4.91]			[4.29]
Anomaly 5					0.11		-0.68
					[3.07]		[-0.11]
Anomaly 6						0.93	0.65
						[7.95]	[5.45]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the ECD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ECD} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.23	0.19	0.14	0.20	0.20	0.14
	[2.48]	[3.03]	[2.54]	[1.79]	[2.67]	[2.58]	[2.00]
Anomaly 1	27.89						14.06
	[7.42]						[3.11]
Anomaly 2		16.34					4.62
		[5.60]					[1.46]
Anomaly 3			32.30				37.49
			[8.10]				[6.51]
Anomaly 4				26.37			11.55
				[6.60]			[2.58]
Anomaly 5					14.28		-8.92
					[3.78]		[-1.69]
Anomaly 6						-1.85	-19.54
						[-0.40]	[-3.94]
mkt	2.30	4.15	2.67	2.80	1.35	1.61	4.53
	[1.30]	[2.25]	[1.52]	[1.57]	[0.74]	[0.88]	[2.57]
smb	1.96	3.51	-0.24	2.61	0.66	0.99	3.46
	[0.77]	[1.32]	[-0.10]	[1.01]	[0.25]	[0.37]	[1.36]
hml	1.68	-3.92	-1.56	4.56	0.53	2.53	-1.73
	[0.49]	[-1.07]	[-0.46]	[1.32]	[0.15]	[0.71]	[-0.49]
rmw	-14.08	-8.49	2.57	-8.42	2.35	1.07	-12.25
	[-3.51]	[-2.18]	[0.75]	[-2.24]	[0.66]	[0.30]	[-2.93]
cma	12.98	16.52	-1.04	17.50	15.87	32.49	8.49
	[2.35]	[2.93]	[-0.16]	[3.24]	[2.48]	[4.28]	[1.12]
umd	1.79	4.37	2.49	1.35	3.41	3.04	0.55
	[1.02]	[2.44]	[1.43]	[0.75]	[1.89]	[1.67]	[0.32]
# months	728	728	732	728	732	732	728
$\bar{R}^2$(%)	16	13	17	15	11	9	23

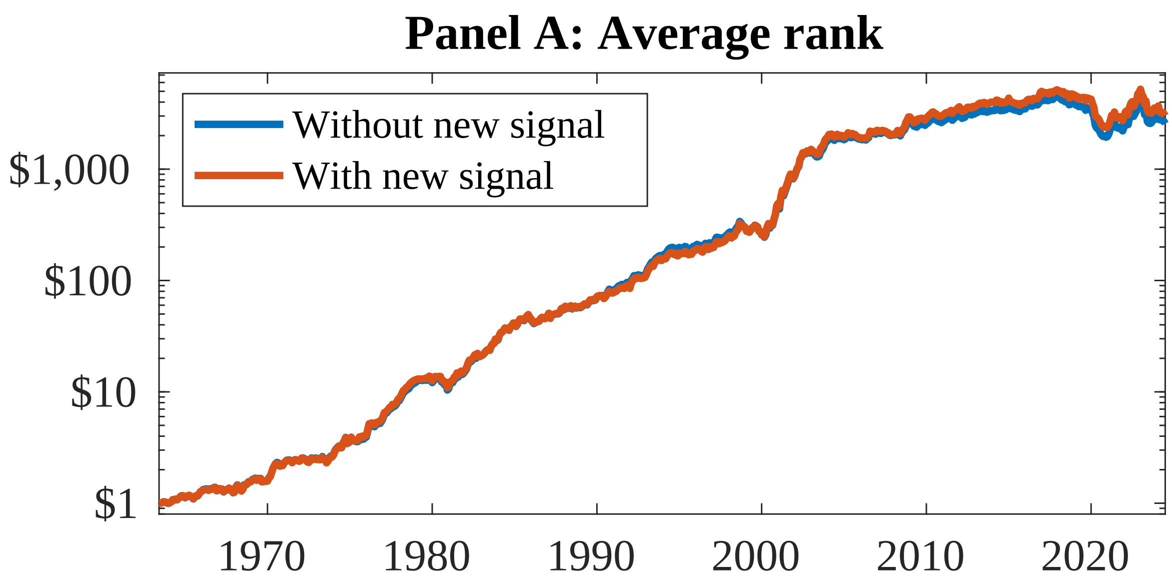


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ECD. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gomes, J. F. (2003). Financing investment. *American Economic Review*, 93(4):1263–1285.

Lyandres, E., Sun, L., and Zhang, L. (2008). The new issues puzzle: Testing the investment-based explanation. *Review of Financial Studies*, 21(6):2825–2855.

Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.

Pontiff, J. and Woodgate, A. (2016). Share issuance and cross-sectional returns: International evidence. *Journal of Financial Economics*, 119(1):44–60.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.

Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.